

411 - Mathematical Analysis - Homework 1

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6. Prove that in a Hausdorff space (hence in every metric space), every finite set is closed.

Solution: Let X be Hausdorff and let $A \subseteq X$ be finite. If $A = \emptyset$, then A is closed. Otherwise, write $A = \{a_1, \dots, a_n\}$. We show that $X \setminus A$ is open. Let $x \in X \setminus A$. For each i , the Hausdorff property gives disjoint open sets U_i and V_i with $x \in U_i$ and $a_i \in V_i$.

The set

$$U = \bigcap_{i=1}^n U_i$$

is open and contains x . Since $a_i \notin U_i$ for every i , U contains no point of A . Thus,

$$x \in U \subseteq X \setminus A.$$

Every point of $X \setminus A$ therefore has an open neighborhood contained in $X \setminus A$. Hence, $X \setminus A$ is open, so A is closed.

8. Let X be a metric space with metric d .

(a) Prove that the following function is also a metric on X :

$$\bar{d}(p, q) := \min\{d(p, q), 1\}.$$

Solution: Let $p, q, r \in X$. We verify the metric axioms.

(i) Non-negativity: Since $d(p, q) \geq 0$, we have $\bar{d}(p, q) = \min\{d(p, q), 1\} \geq 0$.

(ii) Identity of indiscernibles: Since $1 > 0$,

$$\bar{d}(p, q) = 0 \iff d(p, q) = 0 \iff p = q.$$

(iii) Symmetry: Since d is symmetric,

$$\bar{d}(p, q) = \min\{d(p, q), 1\} = \min\{d(q, p), 1\} = \bar{d}(q, p).$$

(iv) Triangle inequality: If $d(p, q) \geq 1$ or $d(q, r) \geq 1$, then

$$\bar{d}(p, r) \leq 1 \leq \bar{d}(p, q) + \bar{d}(q, r).$$

Otherwise, both distances are less than 1, so

$$\bar{d}(p, r) \leq d(p, r) \leq d(p, q) + d(q, r) = \bar{d}(p, q) + \bar{d}(q, r).$$

Therefore, $\bar{d}$ is a metric on X .

(b) Prove that d and $\bar{d}$ induce the same topology; that is, they have the same open sets.

Solution: For every $p \in X$ and $0 < r \leq 1$,

$$\bar{d}(p, q) < r \iff \min\{d(p, q), 1\} < r \iff d(p, q) < r.$$

Therefore,

$$B_{\bar{d}}(p, r) = B_d(p, r).$$

Suppose U is open in (X, d) and $p \in U$. Choose $\epsilon > 0$ such that $B_d(p, \epsilon) \subseteq U$, and let $r = \min\{\epsilon, 1\}$. Then

$$B_{\bar{d}}(p, r) = B_d(p, r) \subseteq B_d(p, \epsilon) \subseteq U.$$

Thus, U is open in $(X, \bar{d})$.

Conversely, suppose U is open in $(X, \bar{d})$ and $p \in U$. Choose $\epsilon > 0$ such that $B_{\bar{d}}(p, \epsilon) \subseteq U$, and let $r = \min\{\epsilon, 1\}$. Then

$$B_d(p, r) = B_{\bar{d}}(p, r) \subseteq B_{\bar{d}}(p, \epsilon) \subseteq U.$$

Thus, U is open in (X, d) .

Hence, the two metrics induce the same topology.

Remark. Although the metrics induce the same topology, they may differ in boundedness: $\bar{d} \leq 1$, even when d is unbounded.

9. Let X be an arbitrary topological space and let $A \subseteq X$. Prove the following.

Remark (Partition Theorem). Every point of X lies in exactly one of the following: the interior of A , the boundary of A , or the interior of $X \setminus A$. Equivalently,

$$X = \text{int}(A) \sqcup \partial A \sqcup \text{int}(X \setminus A).$$

Here, $\sqcup$ denotes a disjoint union.

- (a) A is open if and only if it contains none of its boundary points.

Solution: Suppose A is open. For each $x \in A$, the set A is an open neighborhood of x disjoint from $X \setminus A$. Thus, $x \notin \partial A$, so $A \cap \partial A = \emptyset$.

Conversely, suppose $A \cap \partial A = \emptyset$. If $x \in A$, then x is neither a boundary point of A nor an interior point of $X \setminus A$. By the Partition Theorem, $x \in \text{int}(A)$. Therefore, every point of A is an interior point, so A is open.

- (b) A is closed if and only if it contains all of its boundary points.

Solution: Suppose A is closed. If $x \notin A$, then $X \setminus A$ is an open neighborhood of x disjoint from A . Thus, $x \notin \partial A$. Therefore, $\partial A \subseteq A$.

Conversely, suppose $\partial A \subseteq A$. If $x \in X \setminus A$, then x is neither a boundary point of A nor an interior point of A . By the Partition Theorem, $x \in \text{int}(X \setminus A)$. Thus, $X \setminus A$ is open, so A is closed.

- (c) A is neither open nor closed if and only if it contains some of its boundary points, but not all of them.

Solution: By part (a), A is not open if and only if

$$A \cap \partial A \neq \emptyset.$$

By part (b), A is not closed if and only if

$$\partial A \setminus A \neq \emptyset.$$

Thus, A is neither open nor closed exactly when it contains at least one boundary point and omits at least one boundary point.

10. Let X be an arbitrary topological space. A set is called *clopen* if it is both closed and open. Prove that $A \subseteq X$ is clopen if and only if $\partial A = \emptyset$.

Solution: Suppose A is clopen. By Problem 9(a), $A \cap \partial A = \emptyset$. By Problem 9(b), $\partial A \subseteq A$. Together, these imply $\partial A = \emptyset$.

Conversely, suppose $\partial A = \emptyset$. Then both

$$A \cap \partial A = \emptyset \quad \text{and} \quad \partial A \subseteq A.$$

By Problem 9(a) and (b), A is both open and closed. Hence, A is clopen.